

Simulation of Grover's Algorithm using Coupled Oscillators

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Grover's algorithm is a quantum search algorithm that implements unsorted database search in $O(\sqrt{N})$ time, which is an asymptotically optimal number. Grover's algorithm requires superposition of and interference among a set of basis states which exist as available features in a mechanical system of Coupled Oscillators. We have studied the algorithm in a coupled oscillators system by simulating the time evolution of the numerical solution to the system of partial differential equations that govern the mechanics of this system. We have also studied the amplitude evolution of an array of amplitudes (numbers) when Grover's algorithm is implemented to sort a desired component of the array. We have implemented the binary oracle in Coupled Oscillator system as an elastic reflection of the chosen oscillator. We have considered time evolution of system for half a time period for the diffusion operator in Coupled Oscillator system. Finally we have listed the possible decoherence generating elements that we have not considered in this idealistic simulation and we end with listing a few uses of Grover's algorithm implementation in a Coupled Oscillator system.

I. GROVER'S ALGORITHM

Grover's algorithm is a search algorithm that, given a function's value, finds with high probability the value of the unique input variable that produced the particular value in $O(\sqrt{N})$ order of time.

This scenario is most widely applied to an unsorted database search case, which classically can only be done in $O(N)$ order of time.

An unsorted database search is just a Yes/No query on each of the terms of the database until the required entry is found.

A. Setup

Grover's algorithm is applied on a superposition state with every basis state has equal probability (that is, equal weight) and with time evolution, it evolves the state in which the desired basis state has the highest probability. If $|s\rangle$ denotes the superposition state and $|\omega\rangle$ denotes the target basis state and there exist a total of N basis states then

$$|\langle i|s\rangle|^2 = 1/N, \quad |\langle i|\omega\rangle|^2 = \delta_{ij}$$

The superposition state with equal basis state probability is generated by Hadamard transform of n -qubits initialised with $|0\rangle$ where $n = \log_2 N$.

Thus,

$$H^{\otimes n} |0\rangle = H^{\otimes n} |0\dots 000\rangle = \frac{1}{\sqrt{N}} |0\dots 000\rangle + \frac{1}{\sqrt{N}} |0\dots 001\rangle + \dots + \frac{1}{\sqrt{N}} |1\dots 111\rangle$$

Here, $(H_n)_{i,j} = \frac{1}{2^{\frac{n}{2}}} (-1)^{i \cdot j}$

where $i \cdot j$ is the bitwise dot product of the binary representation of the row and column numbers of the tensor H .

B. Algorithm Steps

The algorithm involves repeated application of two operators, the binary oracle and the diffusion operator, in that order. The binary oracle flips the sign of the target state and the diffusion operator reflects the state about the average, i.e. $|s\rangle$. These rotations are performed alternatively in the two dimensional space and can be visualised as in the following figure 1.

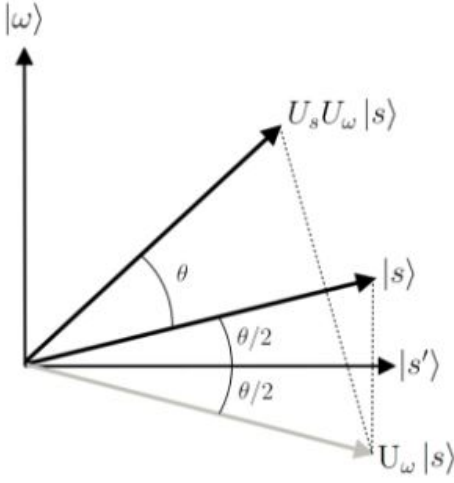


fig.1: Rotations in two dimensional space spanned by $|\omega\rangle$ and $|s\rangle$.

The two reflection operators are the diffusion operator,

$$U_s = 2|s\rangle\langle s| - I$$

and the binary oracle,

$$U_\omega = I - 2|\omega\rangle\langle\omega|$$

c. Simulation

To verify the algorithm, we tested out the algorithm on a vector of size N with $\frac{1}{\sqrt{N}}$ as each of its i th entries. We calculated the matrix of both the operators, binary oracle and diffusion, and repeatedly applied matrix multiplication with the vector, alternatively with both. In nearly $\sqrt{N} * \frac{\pi}{4}$ iterations, the target index's entry gets closer to 1 and all the rest nearly die down to 0. Below is the evolution of target index's entry with number of iterations of the algorithm for $N = 200$.

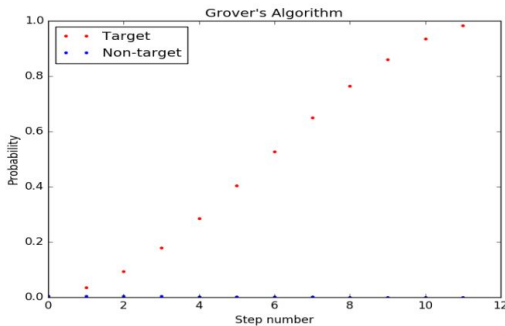


fig.2: A comparison of time required by Grover's algorithm for $N = 200$.

II. IMPLEMENTATION TO HARMONIC OSCILLATOR

There are two necessary conditions to implement this algorithm on any system allowing superposition and interference of states. This is possible in wave dynamics and can be demonstrated in any coupled wave modes provided -

- (1). Superpositioned modes do not undergo phase decoherence.
- (2). Efficient implementation of the two reflection operators is possible.

Here we take a system of N identical harmonic oscillators each representing an item in a database. The oscillators are oscillating in a specific manner while one of them is elastically reflected (the target item in the database). Task at hand is to figure out which one of the target oscillators has been elastically reflected without looking at the process. The optimisation criterion is to design the system of oscillators, and their initial state, so as to make the identification as quick as possible.

To introduce superposition and interference in the system, all the N small oscillators are coupled to a big oscillator (ref.to fig.4) which is then coupled to the center-of-mass mode. This big oscillator coupled to the center-of-mass mode acts as the average state, $|s\rangle$. In this way, the non-target oscillators only influence the target oscillator through the center-of-mass mode and hence are decoupled from the system.

In this setting, elastic reflection of an oscillator implements the binary oracle in velocity space (ref.to fig.3), and time evolution of the whole system by half an oscillation period carries out the reflection about average operation.

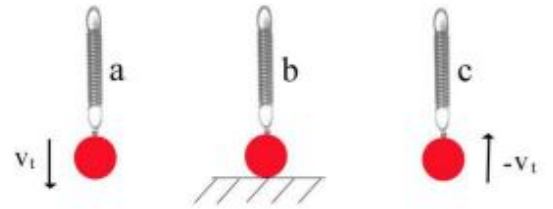


FIG. 3: The binary oracle flips the sign of the target oscillator velocity, when its displacement is zero. (a) Just before the oracle: $x_t = 0$, $v_t > 0$, (b) Elastic reflection: $x_t = 0$, $v_t \rightarrow -v_t$, (c) Just after the oracle: $x_t = 0$, $v_t < 0$.

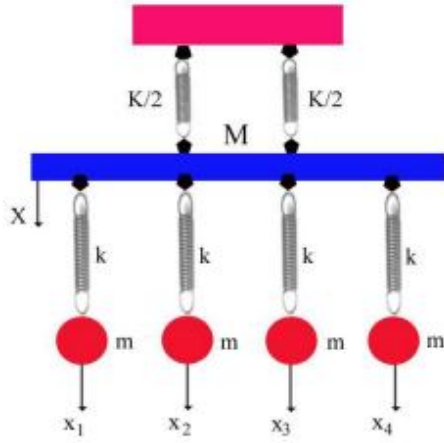


fig.4: A System of four identical harmonic oscillators, coupled to a big oscillator via the center of mass mode.

A. Dynamics

The equations of motions are (with variables carries over from fig.3)-

$$M\ddot{X} = -KX + k \sum_i (x_i - X)$$

$$m\ddot{x}_i = -k(x_i - X)$$

Where $\bar{x}$ is the center-of-mass position. Since non-target oscillators are the decoupled modes, we move to the three variable space $(X, \bar{x}, x_t)$, and our equations change to

$$\begin{aligned} \ddot{X} &= -\frac{K+Nk}{M}X + \frac{Nk}{M}\bar{x} \\ \ddot{\bar{x}} &= -\frac{k}{m}\bar{x} \\ \ddot{x}_t - \ddot{\bar{x}} &= -\frac{k}{m}(x_t - \bar{x}) \implies \omega_t = \sqrt{\frac{k}{m}} \end{aligned}$$

The first two equations are coupled, and their eigenmodes are of the form-

$$X + \lambda \bar{x}$$

$$\begin{aligned} \therefore \ddot{X} + \lambda \ddot{\bar{x}} &= X \left(-\frac{K+Nk}{m} + \lambda \frac{k}{m} \right) + \bar{x} \left(\frac{Nk}{m} - \lambda \frac{k}{m} \right) \\ &= -\omega^2 (X + \lambda \bar{x}) \implies \lambda = \frac{\left(\frac{Nk}{m} - \lambda \frac{k}{m} \right)}{\left(-\frac{K+Nk}{m} + \lambda \frac{k}{m} \right)} \end{aligned}$$

Defining $R_m = \frac{M}{m}$ and $R_k = \frac{K}{k}$, we find sinusoidal solutions with -

$$\omega = \omega_t \sqrt{\frac{N+R_k}{R_m} - \lambda}$$

We have two solutions, ω_+, ω_- , which satisfy,

$$\omega_+^2 + \omega_-^2 = \omega_t^2 \left(1 + \frac{N+R_k}{R_m} \right)$$

$$\omega_+^2 \omega_-^2 = \omega^4 \frac{R_k}{R_m}$$

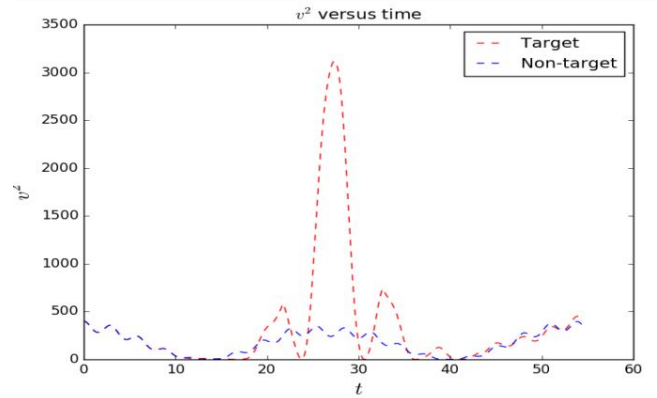
for convenience, we select $\omega_+ = \frac{3}{2}\omega_t$, $\omega_- = \frac{1}{2}\omega_t$, which yield the dynamical equations -

$$x_t = -\frac{1}{2}A_+ \sin(\omega_+ t + \phi_+) + \frac{1}{2}A_- \sin(\omega_- t + \phi_-) + A_t \sin(\omega_t t + \phi_t)$$

$$x_{i \neq t} = -\frac{1}{2}A_+ \sin(\omega_+ t + \phi_+) + \frac{1}{2}A_- \sin(\omega_- t + \phi_-) - \frac{1}{N-1}A_t \sin(\omega_t t + \phi_t)$$

B. Simulations

We simulated the time evolution of the dynamical equations of this system. The dynamical equations were obtained by numerical integration of the system of Partial Differential Equations obtained analytically above. We solved the system for $N=100$. Fig. 5 depicts the time evolution of the square of velocity for target and non-target harmonic oscillators.



C. References

1. Aavishkar A. Patel, A Coupled Oscillator Model for Grover's Quantum Database Search Algorithm, arXiv:0711.4733 [physics.gen-ph]
2. L.K. Grover, A Fast Quantum Mechanical Algorithm for Database Search, arXiv.org:quant-ph/9605043.